Michael **Mrozek**

Software Engineer

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Technical Skills

Languages and Frameworks

- Extensive experience with C, Python, Typescript, and scripting.
- Experience with C++, Java, and Rust.
- Experience with SQL and miscellaneous databases, including MySQL, MariaDB, PostgreSQL, and Redis.
- Experience with React, Vue, Vite, Webpack, Yarn, and other modern web development tooling.

Tools

- Extensive experience with Linux and Windows.
- Extensive experience with Git. Some experience with Subversion.
- Experience with VSCode, Emacs, PyCharm, and other editors/IDEs.
- Experience with the Atlassian toolset, including Jira, Bitbucket, and Confluence.
- Personal experience with Firebase and cloud hosting.

Experience

Owl Cyber Defense *2024 - Present*

Principal Software Engineer

- Implemented new protocol adapters and features in a commercial data diode and a government CDS.
- Automated common build and test procedures.

Mercury *2016 - 2024*

Senior Principal Software Engineer

- Designed and rolled out an automated build and test system for a number of projects.
- Handled the office migration to Atlassian tools, including moving all projects from Subversion to Bitbucket.
  - Implemented a tool to manage cross-repository dependencies in Bitbucket, and an addon to track additional project information within Bitbucket.
- Fully automated a recurring task needed by a customer, so it can now be done on demand in about 90 minutes.
- Fixed and redesigned large portions of an embedded system of interconnected Microblaze cores used by many customers.
- Handled communications for several programs, including answering customer questions, debugging issues, and releasing new versions.
- Assisted newer engineers with understanding tasks, solving problems, debugging issues, and generally answering questions about the system architecture.
- Approved pull requests before they could be merged and suggested fixes or improvements.

Microsemi *2010 - 2016*

Software Developer

- Implemented and improved miscellaneous whitebox cryptography algorithms, including RSA, AES, and elliptic curve DSA.
- Developed an OpenSSL engine wrapping our whitebox cryptography implementations so they could be used by any application utilizing OpenSSL.
- Implemented the security model and key management policy for an Android password manager.
- Designed and implemented multiple higher level protocols wrapping existing whitebox cryptography to satisfy customer needs.
- Red- and blue-teamed several devices, finding a number of vulnerabilities. Wrote large portions of the final reports detailing the vulnerabilities, their severity, and potential mitigations and fixes.

Education

Purdue University *West Lafayette, IN*

MS Computer Science *May 2015*

Rose-Hulman Institute of Technology *Terre Haute, IN*

BS Computer Science, Software Engineering (Cum Laude) *May 2009*

Projects

Serial Bridge

A web-based remote serial port interface heavily utilized at Mercury to interact with hardware UARTs.

Chalk

A coding interview site with a collaborative live code editor, question library, and compiler.

Point Vote

A website for hosting live scrum planning poker sessions and other votes.

Ninja

A website for performing calculations via a graphical data flow.

Sprint

A web-based Scrum tracking tool tailor-made for our project needs at Microsemi.

Noisebot

An IRC/Slack chat bot that supports reloadable modules and easy modification by end-users.